Input: A protocol Π

Output: A rectangle $R = (\{\sigma_1\} \times X_1) \times \cdots \times (\{\sigma_k\} \times X_k)$ and k sets $J_1, \ldots, J_k \subseteq [n]$.

for $i \in [k]$ **do**
  $X_i := \{0,1\}^n, J_i := [n]$. `// Initialization`
Sample $\sigma_1 \sim [n]$.
$v :=$ root of Π, $R := (\{\sigma_1\} \times \{0,1\}^n) \times ([n] \times \{0,1\}^n)^{k-1}$, bad := FALSE.
while *v is not a leaf node* **do**
  $i := \mathsf{owner}(v), u_0 := \Pi(v,0), u_1 := \Pi(v,1)$.
  `//Loop invariant: (1)` $R \subseteq \Pi_v$`; (2)` $X_i(J_i)$ `is` γ`-dense.`
  X_i is partitioned into $X_i = X^0 \cup X^1$ according to Π.
  Sample $\boldsymbol{b}$ such that $\mathbf{Pr}[\boldsymbol{b} = b] = \frac{\chi_k(R^b)}{\chi_k(R)}$ where $R^b = (\{\sigma_1\} \times X_1) \times \cdots (\{\sigma_i\} \times X^b) \times ([n] \times \{0,1\}^n)^{k-i}$ for $b \in \{0,1\}$.
  `//`R `is always a shorthand for` $(\{\sigma_1\} \times X_1) \times \cdots \times (\{\sigma_i\} \times X_i) \times ([n] \times \{0,1\}^n)^{k-i}$
  Update $X_i := X^{\boldsymbol{b}}$.
  Let $X_i = X^1 \cup \cdots \cup X^m$ be the decomposition of X_i promised by Lemma 13 with associated sets $I_1, \ldots, I_m \subseteq J_i$. `// Invoking Lemma 13 with` $\gamma = 1 - \frac{2\epsilon}{k}, J = J_i, N = 2$.
  Sample $\boldsymbol{j} \in [m]$ such that $\mathbf{Pr}[\boldsymbol{j}' = j] = \frac{\chi_k(R^j)}{\chi_k(R)}$ where $R^j = (\{\sigma_1\} \times X_1) \times \cdots \times (\{\sigma_i\} \times X^j) \times ([n] \times \{0,1\}^n)^{k-i}$ for $j \in [m]$.
  Update $X_i := X^j, J_i := J_i \setminus I_j$.
  if $\mathsf{owner}(u_{\boldsymbol{b}}) \neq i$ **then**
    Sample $\sigma_{i+1} \in [n]$ such that $\mathbf{Pr}[\sigma_{i+1} = \rho] = \frac{\chi_k(R^\rho)}{\chi_k(R)}$ where $R^\rho = (\{\sigma_1\} \times X_1) \times \cdots \times (\{\sigma_i\} \times X_i) \times (\{\rho\} \times \{0,1\}^n) \times ([n] \times \{0,1\}^n)^{k-i-1}$ for $\rho \in [n]$.
    if $\sigma_{i+1} \notin J_i$ **then** bad := TRUE;
  Output $R = (\{\sigma_1\} \times X_1) \times \cdots \times (\{\sigma_k\} \times X_k), J_1, \ldots, J_k$.